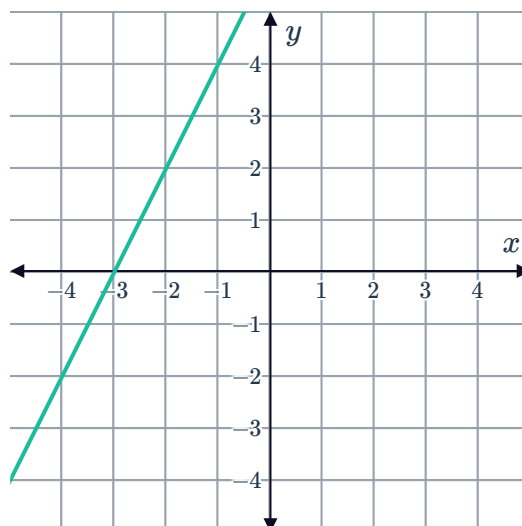


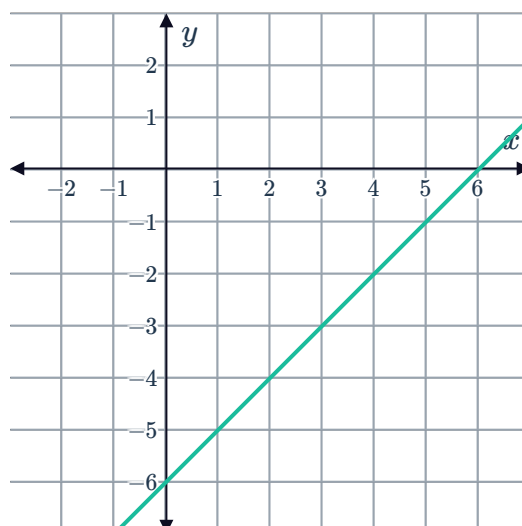
Linear equations and inequalities



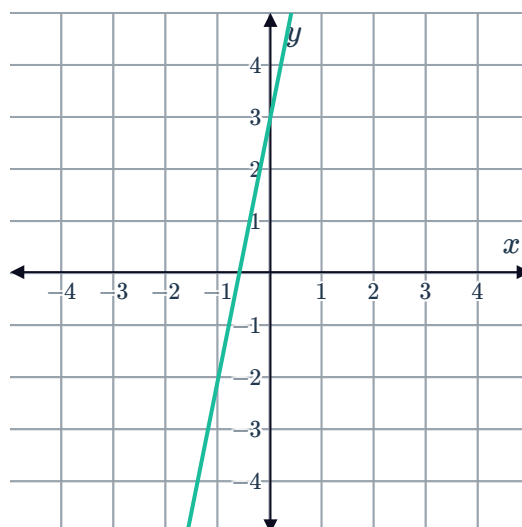
- 1 Consider the graph of $y = 2x + 6$. Using the graph, solve the inequality $2x + 6 \geq 0$.



- 2 Consider the graph of $y = x - 6$. Using the graph, solve the inequality $x - 6 < 0$.



- 3 Consider the graph of $y = 5x + 3$. Using the graph, solve the inequality $5x + 3 \leq 0$.

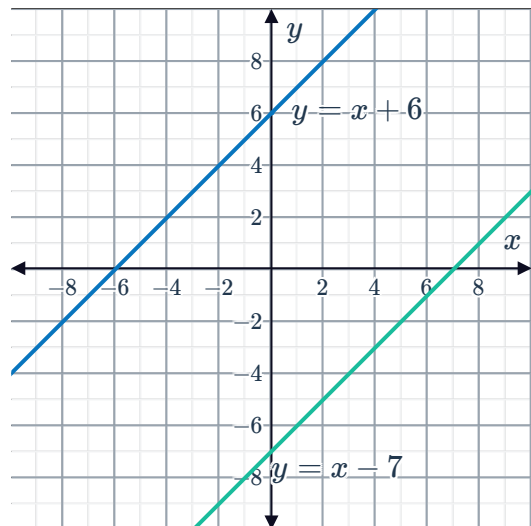


4 Consider the equation $2(x - 1) - 3 = 7$.



- a Solve for the value of x that satisfies the equation.
- b To verify the solution graphically, what two straight lines need to be graphed?
- c Graph these lines on the same number plane.
- d Hence find the value of x that satisfies the two equations.

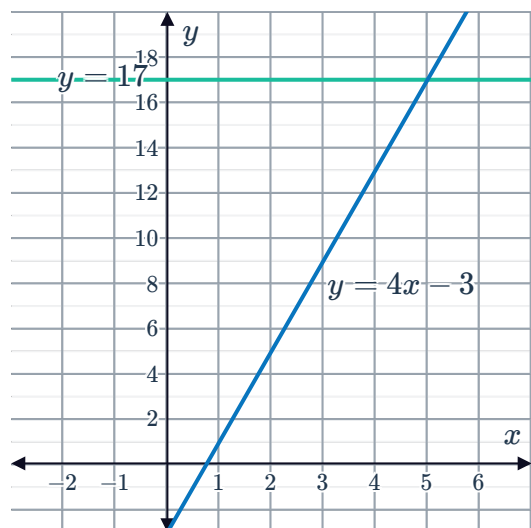
5 Consider the graphs of $y = x + 6$ and $y = x - 7$:
How many solutions does the inequality $x + 6 \geq x - 7$ have?



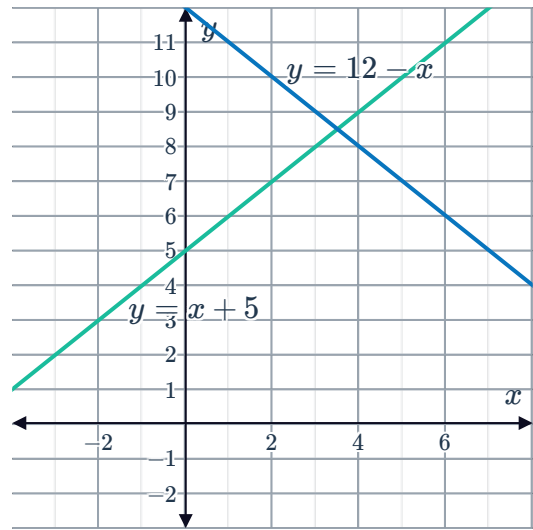
6 Consider the inequality $2x - 4 > 2 - 4x$.

- a Sketch the graphs of the lines for $y = 2x - 4$ and $y = 2 - 4x$.
- b Find the point of intersection of the lines.
- c Hence, solve the inequality $2x - 4 > 2 - 4x$.

7 Consider the graph of the lines $y = 17$ and $y = 4x - 3$:
Using the graphs, solve the inequality $4x - 3 < 17$.



- 8 Consider the graphs of $y = x + 5$ and $y = 12 - x$:
Using the graphs, solve the inequality $x + 5 > 12 - x$.



- 9 Use sign diagrams to solve the following inequalities:

a $5x - 15 > 0$

b $8x + 32 \leq 0$

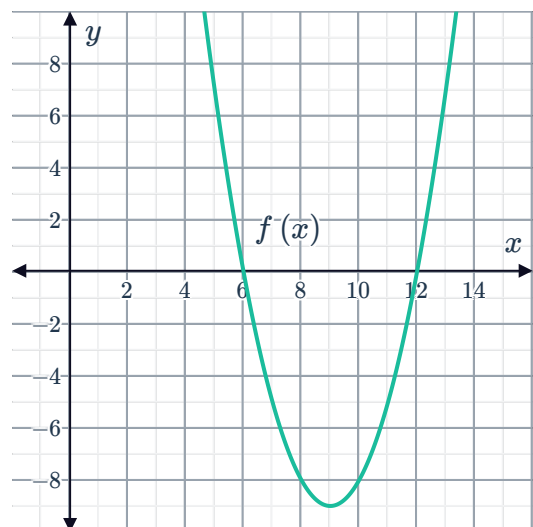
c $-3x + 81 \geq 0$

d $-20x > 80$

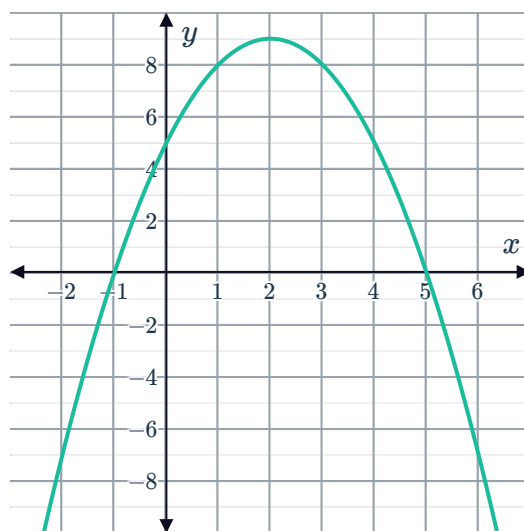
Quadratic equations and inequalities

- 10 Consider the graph of $y = f(x)$:

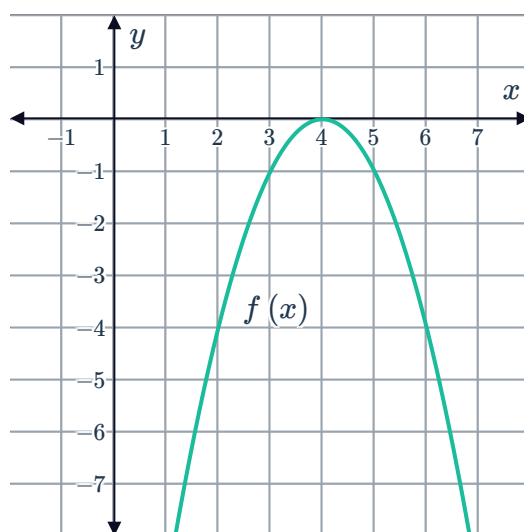
- Find the values of x for which $f(x) = 0$.
- For what values of x is $f(x) < 0$?
- For what values of x is $f(x) > 0$?
- What is the x -coordinate of the vertex of $f(x)$?



- 11 Consider the function $f(x) = 5 + 4x - x^2$.
Use the graph to solve the inequality $5 + 4x - x^2 > 0$.



- 12 Consider the graph of $y = f(x)$:
- For what values of x is $f(x) < 0$?
 - For what value of x is $f(x) \geq 0$?
 - What is the axis of symmetry of $f(x)$?
 - What is the value of the discriminant of $f(x)$?



- 13 Consider the function $f(x) = x^2 - 4x - 5$.
- Sketch the graph of the function.
 - Hence state the values of x for which $f(x) \leq 0$.

- 14 Consider the function $f(x) = 3x^2 - 2x - 8$.
- Solve the equation $f(x) = 0$.
 - Sketch the graph of the function.
 - Hence state the values of x for which $f(x) \geq 0$.

- 15 Consider the inequality $(x - 3)^2 \leq 0$.
- How many x -intercepts does the graph of $y = (x - 3)^2$ have?
 - Solve the inequality.

16 Consider the function $y = 2x^2 + 9x + 8$.



- a Determine the x -intercepts of the function.
- b Is the graph concave up or concave down?
- c Hence find the values of x for which $y > 0$.

17 Consider the function $f(x) = x^2 - 2x$.

- a Sketch the graph of the function.
- b Hence state the values of x for which $f(x) \leq 8$.

18 Consider the inequality $x^2 - 2x \leq -x + 2$.

- a Sketch the graphs of $y = x^2 - 2x$ and $y = -x + 2$ on the same number plane.
- b State the x -values for the points of intersection.
- c Hence solve the inequality $x^2 - 2x \leq -x + 2$.

19 Consider the inequality $x^2 > 6x - 5$.

- a Sketch the graphs of $y = x^2$ and $y = 6x - 5$ on the same number plane.
- b State the x -values for the points of intersection.
- c Hence solve the inequality $x^2 > 6x - 5$.

20 Consider the inequality $3x^2 + x \geq 2x^2 + 2$.

- a Sketch the graphs of $y = 3x^2 + x$ and $y = 2x^2 + 2$ on the same number plane.
- b Hence or otherwise, solve the inequality $3x^2 + x \geq 2x^2 + 2$.

21 Use sign diagrams to solve the following inequalities:

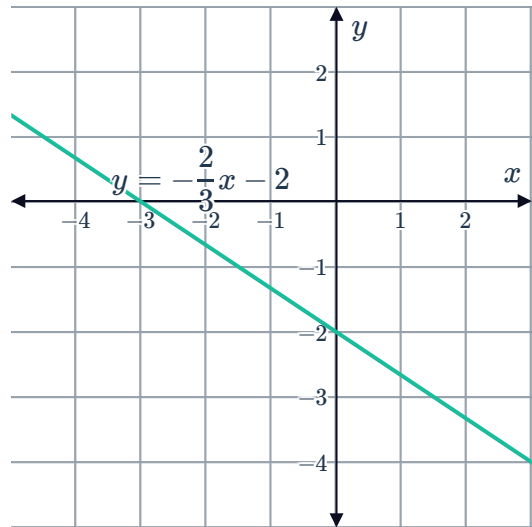
- | | |
|-------------------------|-----------------------|
| a $x^2 - 9 > 0$ | b $2x^2 - 4x \leq 0$ |
| c $x^2 - 5x + 6 \geq 0$ | d $x^2 + 7x + 12 > 0$ |

Equivalent inequalities

22 To solve the inequality $x \leq 2x - 3$, Christa graphed $y = x + 3$. What other line would she need to graph to be able to solve the inequality graphically?

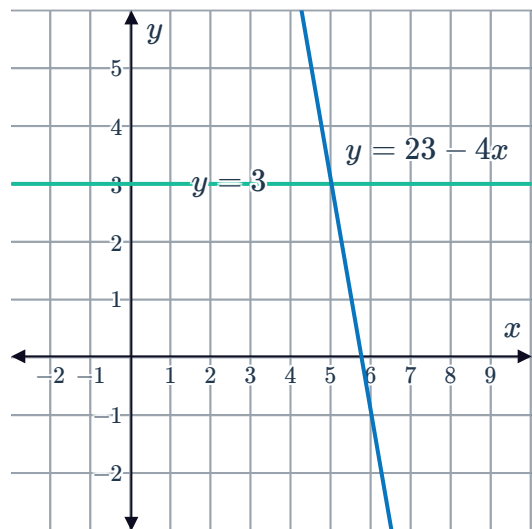
23 To solve the inequality $x \leq \frac{x-3}{4} - 1$, Track graphed $y = x - 3$. What other line would she need to graph to be able to solve the inequality graphically?

24 Consider the graph of $y = -\frac{2}{3}x - 2$.
Using the graphs, solve the inequality $-2x - 6 > 0$.

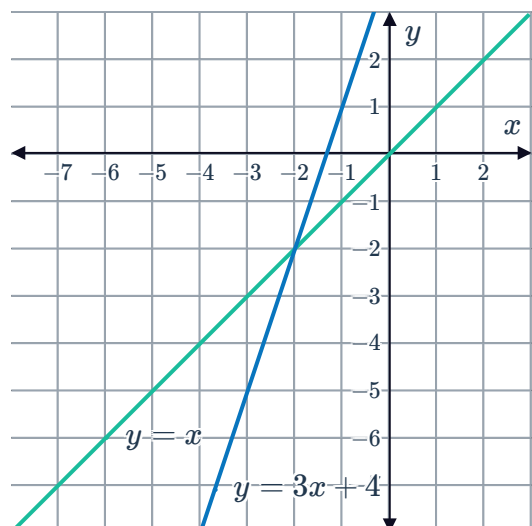


25 Consider the graph of the lines $y = 3$ and $y = 23 - 4x$:

- a** Using the graphs, solve the inequality $23 - 4x < 3$.
- b** Using the graphs, solve the inequality $-20 + 4x \geq 0$.



26 Consider the graphs of $y = 3x + 4$ and $y = x$.
Using the graphs, solve the inequality $3x + 4 - x \leq 0$.



27 Explain whether the following inequalities can be solved graphically using the graphs of:

$$f(x) = 4x + 3 \text{ and } g(x) = 5 - x$$

a $(4x + 3) - (5 - x) \leq 0$

b $(4x + 3) + (5 - x) < 0$

c $\frac{4x + 3}{5} + x > 0$

d $3x + 8 < 0$

e $5x > 2$